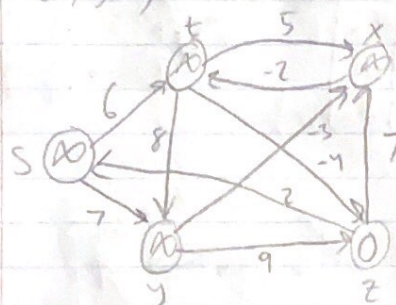


Parth Patel
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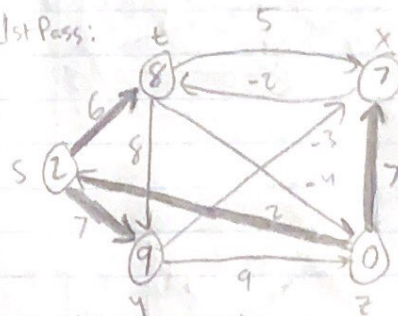
10/23/23
Problem Set 3

22.1-1 Run the Bellman-Ford algorithm on the directed graph of Figure 22.4, using vertex z as the source. In each pass, relax edges in the same order as in the figure, and show the d and π values after each pass. Now, change the weight of edge (z, x) to 4 and run the algorithm again, using s as the source. (Figure 22.4 says that each pass relaxes the edges in the order of: (t, x) , (t, y) , (t, z) , (x, t) , (y, x) , (y, z) , (z, x) , (z, s) , (s, t) , (s, y))



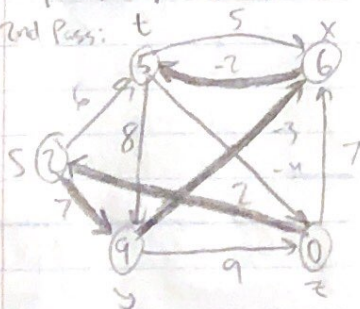
	s	t	x	y	z
d	∞	∞	∞	∞	0
π	NIL	NIL	NIL	NIL	NIL

1st Pass:



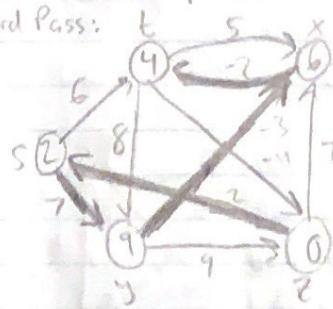
	s	t	x	y	z
d	2	8	7	9	0
π	z	s	z	s	NIL

2nd Pass:



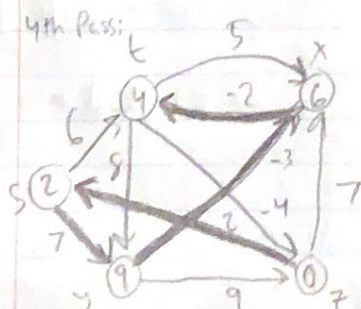
	d	π
s	2	z
t	5	x
x	6	y
y	9	s
z	0	NIL

3rd Pass:



	d	π
s	2	z
t	4	x
x	6	y
y	9	s
z	0	NIL

4th Pass:



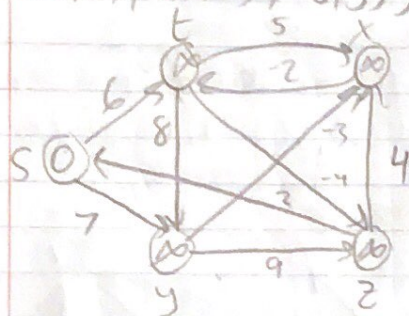
	d	π
s	2	z
t	4	x
x	6	y
y	9	s
z	0	NIL

No negative cycles, so this algorithm will return true.

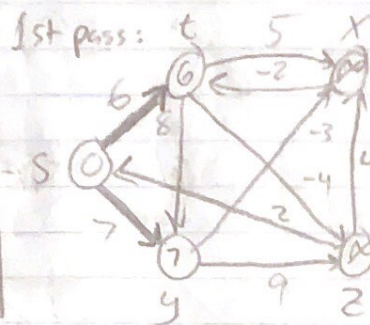
The thick lines are for the predecessors

22.1-1 (continued)

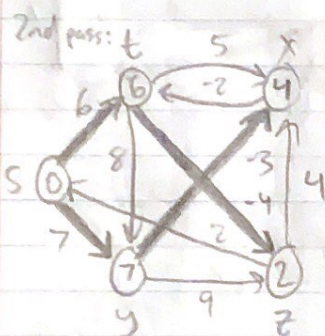
Let the weight of edge (z,x) equal 4 and let's run Bellman-Ford algorithm again using "S" as the source. (Figure 22.4 says that each pass relaxes the edges in the order of: $(t,x), (t,y), (t,z), (x,t), (y,x), (y,z), (z,x), (z,s), (s,t), (s,y)$)



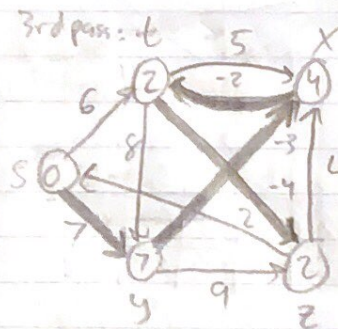
	d	π
S	0	NIL
t	∞	NIL
x	∞	NIL
y	∞	NIL
z	∞	NIL



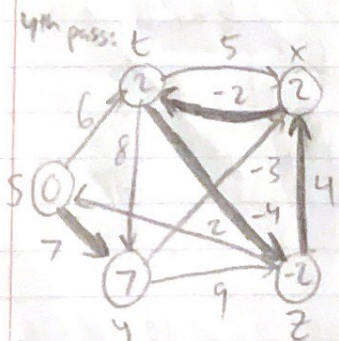
	d	π
S	0	NIL
t	6	S
x	6	NIL
y	7	S
z	∞	NIL



	d	π
S	0	NIL
t	6	S
x	4	y
y	7	S
z	2	t



	d	π
S	0	NIL
t	2	x
x	4	y
y	7	S
z	2	t

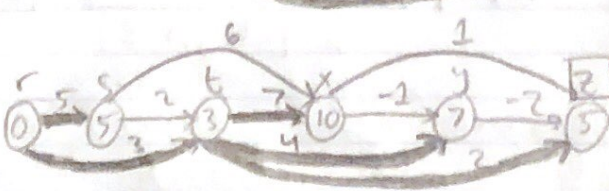
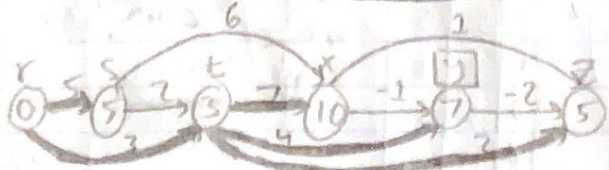
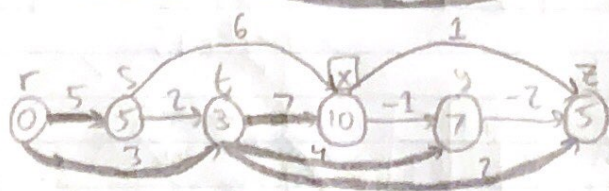
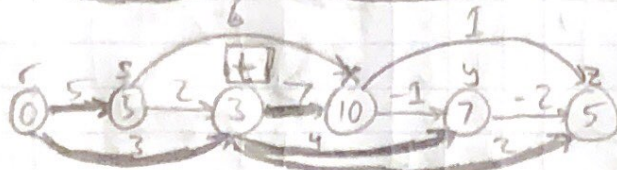
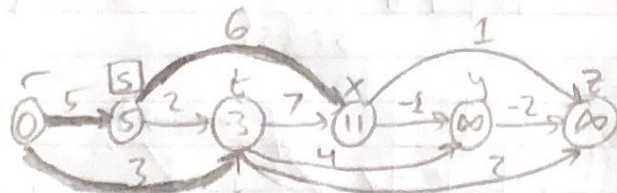
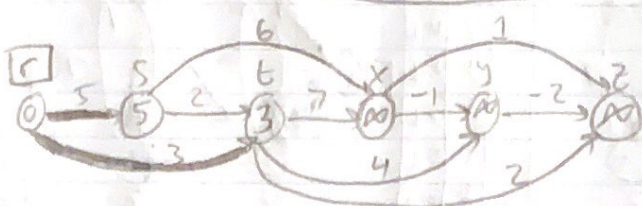
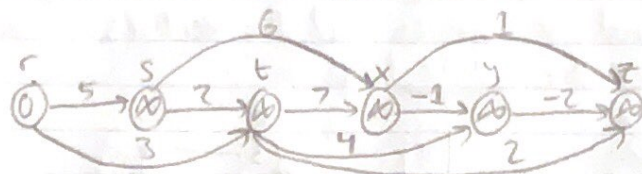


	d	π
S	0	NIL
t	2	x
x	2	t
y	7	S
z	-2	t

The thick lines on the graphs are for the predecessors

This algorithm will return false, since there is a negative weight cycle between the vertices: $\{z, x, t\}$. Since Bellman-Ford has a second for-loop to check for negative cycles, I noticed that for edge (x,t) , $t.d$ was not greater than $x.d + w(x,t)$: $t.d = 2$, $x.d = 2$, $w(x,t) = -2$, and $2 > 2 + (-2)$ is false, so this algorithm would return false.

22.2-1 Show the result of running DAG-SHORTEST-PATHS on the directed acyclic graph of Figure 22.5, using vertex "r" as the source.



★ I show the vertex that I was on by putting a box: $\square$, around it
★ The thick lines are for the predecessors for the shortest path algorithm